

ABHIJIT SWAIN

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PROFESSIONAL SUMMARY

Computer Applications graduate with strong fundamentals in C/C++, object-oriented design, and data structures, applied through hands-on full-stack and real-time systems development (JavaScript/TypeScript, Node.js, React, Python). Experienced building event-driven, concurrent applications (WebRTC/Socket.IO) with attention to performance, modular architecture, and clean SDLC practices. Actively building applied Generative AI and Agentic AI skills — RAG pipelines, LLM integration, and the Model Context Protocol (MCP) — alongside core full-stack engineering. Comfortable working close to the system layer (Linux environments, Docker, Git-based CI/CD) and quick to pick up new platforms and protocols. Open-source contributor with a collaborative, problem-solving approach to engineering.

TECHNICAL SKILLS

Languages	C++, C, JavaScript (ES6+), TypeScript, Python
AI / GenAI	Generative AI, Agentic AI, RAG (Retrieval-Augmented Generation), LLM integration, Model Context Protocol (MCP)
CS Fundamentals	Object-Oriented Programming, Data Structures & Algorithms, Operating System Concepts, Multithreading/Concurrency Basics
Systems & Platforms	Linux environments, Git version control, Docker, Kubernetes (EKS), CI/CD (GitHub Actions, GitLab CI, CircleCI)
Real-Time & Networking	WebRTC, Socket.IO, RESTful API design, event-driven architecture, microservices
Web & Backend	React.js, Node.js, Express.js, MongoDB, PostgreSQL, RBAC/JWT authentication
Cloud & Monitoring	AWS (EC2, S3, Lambda, API Gateway), Prometheus, Grafana, ELK Stack

KEY PROJECTS

EchoMeet — Real-Time Video Conferencing Platform	2025
- Engineered a peer-to-peer real-time video conferencing system using WebRTC and Socket.IO with Node.js/Express, handling multiple concurrent sessions through an event-driven, low-latency architecture. - Designed session-management and signaling logic in C-like modular fashion (clear state machines for call setup, teardown, and reconnection), applying OOP and data-structure fundamentals to manage concurrent client state. - Built support for screen sharing and in-app messaging; structured the codebase for maintainability with Git-based version control and CI/CD-aligned delivery workflows.	
WanderLust — Full-Stack Property Listing Platform	2024
- Developed a full-stack MERN application with secure JWT-based authentication and role-based access control (RBAC), emphasizing clean OOP design and RESTful API principles. - Implemented dynamic CRUD operations with robust frontend-backend integration; containerized with Docker and deployed via a GitHub Actions CI/CD pipeline. - Maintained code quality through unit testing, debugging cycles, and modular architecture across iterative feature development.	
Anigram — Open Source Contribution — GitHub	2024
- Identified and resolved critical bugs to improve application stability; implemented new frontend rendering features collaborating with a distributed team via pull requests and code review. - Documented recurring error patterns and reusable resolution approaches, contributing to long-term codebase maintainability.	

EDUCATION

Master of Computer Applications (MCA)	May 2026
Srusti Academy of Management And Technology, Bhubaneswar	
Bachelor of Computer Applications (BCA)	May 2024
Regional College Of Management, Bhubaneswar	

PROFESSIONAL DEVELOPMENT

- Software Engineering Fundamentals** — OOP, data structures & algorithms, SDLC best practices, applied across full-stack and real-time systems projects.
- Applied AI & Agentic Systems** — Python, Generative AI, Agentic AI, RAG pipelines, LLM integration, and the Model Context Protocol (MCP).
- Full Stack & Cloud** — MERN stack, RESTful APIs, Docker, Kubernetes fundamentals, CI/CD (GitHub Actions), AWS basics (EC2, S3, Lambda).
- Open Source Contributor** — Active GitHub contributor with experience in distributed workflows, PR reviews, and issue resolution.